Reflection and Traceability Report on Physiotherapy Companion

Team 11, technically functional
Matthew Huynh
Cieran Diebolt
Vaisnavi Shanthamoorthy
Maham Siddiqui
Eman Ashraf

Contents

List of Tables

List of Figures

1 Symbols, Abbreviations and Acronyms

symbol	**description**
FR	Functional Requirement
NFR	Non-Functional Requirement
HA	Hazard Analysis
MG	Module Guide
MIS	Module Interface Specification
SRS	Software Requirement Specification
TA	Teaching Assistant
VnV	Verification and Validation

2 Changes in Response to Feedback

2.1 SRS and Hazard Analysis

The hazard analysis document was improved through the peer review along with TA feedback in several areas for enhancement, including clarifying the roadmap with implementation timelines and rationales for safety requirements (#144), adding justification for future work prioritization (#150), and removing problematic assumptions about user familiarity with smart devices (#136). Inconsistencies between the SRS and Hazard Analysis were resolved (#129), while the system boundaries and components section was expanded with lower-level explanations to remove ambiguity (#135). The FMEA table was further decomposed to represent detailed failure modes rather than general effects (#142), and hazard categorization was clarified for technical hazards that could impact safety (#153). Additional improvements included adding rationales for each critical assumption (#136), addressing inconsistencies between assumptions and failure modes (#143), and noting where mitigation strategies would be expanded (#152). TA feedback further reinforced the need for roadmap rationales, which was addressed alongside similar peer feedback. One comment regarding bibliography completeness (#151) was determined to be invalid as the citation and reference were already properly included.

Source	**Issue #**	**Issue Description**	**Changes Made**	**Commit**
Peer Review	#151	Citation "IppolitoWallace1995" appears but bibliography incomplete	No change - comment invalid as bibliography and in-text citation were complete	None
Peer Review (Group 14)	#144	Unclear description of how/when safety requirements and hazard testing will be implemented	Added section in roadmap outlining rationale for accepting requirements within capstone timeline and rationales for excluded safety requirements	939b502

Peer Review (Group 14)	#150	Safety requirements prioritized as "first" vs "in the future" without clear justification or timeline	Added clear rationale for future work prioritization	470F490
Peer Review (Group 14)	#136	Missing rationale for each assumption; assumption about user familiarity with smart device should be avoided	Removed assumption about user familiarity; added rationales for each assumption	a103812
Peer Review (Group 14)	#143	Inconsistency between assumption about user familiarity and "confusing prompts" failure mode	Removed the initial assumption; will update language of "confusing prompts"	None (different commit)
Peer Review (Group 14)	#129	Inconsistencies between SRS and Hazard Analysis; hazards need prioritization by severity	Updated consistency between SRS and HA	Reference #410 PR
Peer Review (Group 14)	#135	System components need lower-level explanation with sample hazards; authentication module lacks detail	Added changes to HA system boundaries, splitting section to remove ambiguity	4bd898a
Peer Review (Group 14)	#142	FMEA rows should represent detailed failure modes rather than effects	Included further decomposition of table within FMEA	Different commit

Peer Review (Group 14)	#153	Technical hazards could also be safety hazards leading to unsafe exercise	Clarified hazard categorization	4381828
Peer Review (Group 14)	#152	Document lacks detailed mitigation strategies and implementation specifics	Mitigation included; will add section about effectiveness	None (different section)
TA	N/A	For Section 8 (Roadmap), including rationale would be helpful for which safety requirements to implement and why they are in that time frame	Updated the section with timelines and rationale for why each safety requirement was placed in each term	Similar to #144

Table 1: Feedback and changes for SRS and Hazard Analysis

2.2 Design and Design Documentation

In reference to changed made between Rev0 and Rev1, the MG was modified to reflect changes in the overall module design of the project. The MG modules were modified in their responsibilities and locations in the overall design (Refer to Issue #408). There were changes made to not only the modules themselves but the justification for anticipated changes, unlikely changes, and design decisions to further explain the effect they held on the final design of the project. It is hoped this additional level of explanation will help to clarify some of the limitations, assumptions, and constraints that were presented in several of the modules and other documents (Refer to PR #416). Changes to the modules required changes to the uses hierarchy and the traceability sections also found in the MG. Finally, there were modifications to the MG's timelien section to make it more focussed on modules and design documentation with specific dates rather than overall deliverables and general date ranges (Refer to Issue #407). Further changes are documented below in Table 2.

Source	**Issue #**	**Issue Description**	**Changes Made**	**Commit**

Peer Review (Group 14)	#242	Hardware Hiding Module reference issues and Modules lacking detail	Hardware Hiding Module reference fixed, however module details did not seem to be lacking as per team and TA feedback; no change on this part.	7edbcd4
Peer Review (Group 14)	#241	Module names unclear	Module names updated for clarity. Home changed to Profile Activity, Main changed to Analysis Control Flow.	d2b2239

Peer Review (Group 14)	#244	Home, Main, and Database modules seem multi-purposed. Consider rewording secrets or fracturing modules into several new ones.	The Main module was renamed to the Analysis Control Flow module and handles the overall logic for recording, so was unchanged. The Profile Activity module was updated to reflect the new responsibilities, then eventually scrapped as the UI was redesigned and absorbed the responsibilities. The Database module was decomposed into the Account Creation, Valid User, User Account, User Activity, and Exercise Data modules which helped to split secrets and make modules clearer.	adefa15

TA	#407	Timeline done at deliverable level rather than module implementation level and ambiguous team member assignments.	Timeline updated to be more specifically focused on module implementation with specific target dates and team member assignments. Design documentation also included in the timeline, but other deliverables were removed.	bef9394
TA	#408	ACs, UCs, and Design Decisions lacking information.	ACs, UCs, and Design Decisions updated with more rationale and impact on the overall design.	8648a1c

Table 2: Feedback and changes for MIS

2.2.1 MIS

While the feedback received from the TA about the content was positive, there were a lot of formatting errors present in the original document. There were large spaces, inconsistent headings and sections marked as 'TBD' and 'TODO'. These sections' entries are in progress and all the issues relayed in the TA feedback (#370) are being resolved. Peer feedback issues were very similar (#243) with the addition of further description being needed for each module(#240) and clarifying assumptions (#245)

Source	**Issue #**	**Issue Description**	**Changes Made**	**Commit**
TA feedback	#370	Inconsistent formatting and empty entries present	This is a work in progress	None

Peer Review (Group 5)	#243	Sections are empty with 'TODO' and 'TBD' labels	Added consistent headings and some completed sections	b81182e, 826d3b1, 64c8c8c, 11d112a, c9b8fc0, 030133a, 6c02172, 680a1b1, c35a5b1
Peer Review (Group 5)	#240	UI section incomplete and lack of description under each module	Adding further descriptions and UI module is in progress	None
Peer Review (Group 5)	#245	Same as #240 and #243; Assumptions not present even thought they are being made	Some assumptions fixed, others are works in progress	b81182e, 826d3b1, 64c8c8c, 11d112a, c9b8fc0, 030133a, 6c02172, 680a1b1, c35a5b1

Table 3: Feedback and changes for MIS

2.3 VnV Plan and Report

In terms of the Verification and Validation (V&V) Plan, the feedback was applied based on both the peer review and TA feedback. The main changes that have addressed include adding clear pass/fail criteria for test cases (#167), replacing vague script-based testing descriptions with detailed explanations of implemented unit tests (#168, #428, #429), and improving traceability between requirements and test cases (#172).

Source	**Issue #**	**Issue Description**	**Changes Made**	**Commit**
Peer Review	#167	Add a clear pass/fail criteria	Added clear pass/fail criteria	c8aa78e

Peer Review	#168	Script-based testing lacks detail on how tests are performed	Removed script-based testing and replaced with detailed descriptions of implemented unit test cases, documented individually in the V&V Report	c8aa78e
Peer Review	#169	Fix revision date history	Updated revision date history	cf936e1
Peer Review	#170	Symbols, abbreviations, and acronyms table lacks detail	Expanded and improved table detail	c8aa78e
Peer Review	#171	The document references external LaTeX input files on lines 18-19 that may not exist or be properly linked in the repository structure	No change made	None
Peer Review	#172	Lack of traceability between requirements and unit tests	Revised to include the correct traceability mapping between both	c8aa78e
Peer Review	#173	Unclear distinction between system testing and functional testing sections	Updated the respective sections and added introductions to specify what will be covered in each section	8912684

Peer Review	#174	Explanations for relevance of documentation are vague	Increased level of detail in explanations	c8aa78e
Peer Review	#175	Lack of detail in the Verification and Validation Plan section	Expanded and clarified section content	c8aa78e
TA Feedback	#414	Table formatting could be improved	Cleaned and improved table formatting	c8aa78e
TA Feedback	#427	Missing detail in automated testing and implementation verification	Added more specific detail to these sections	c8aa78e
TA Feedback	#428	Script-based testing descriptions lack clarity on execution	Replaced script-based testing with detailed descriptions of all of the implemented unit test cases. Each of these test cases are documented individually in the V&V Report	c8aa78e
TA Feedback	#429	Ambiguity in scripting for functional testing	Replaced script-based testing with detailed descriptions of all of the implemented unit test cases. Each of these test cases are documented individually in the V&V Report	c8aa78e

TA Feedback	#430	Missing detail in code walkthroughs and static testing	Added additional detail to these sections	c8aa78e
TA Feedback	#431	Missing reference in-text citation in the Objectives section	Added the missing reference	c8aa78e

Table 4: Feedback and changes for Verification and Validation Plan

In terms of the Verification and Validation (V&V) Report, the feedback was applied based on the TA feedback in several sections. There were no peer review issues found for this document, so we based it solely on the TA feedback provided through Avenue. Some of the changes that we applied included improving the level of detail when comparing the expected vs. actual result of the outputs (#432), fixing the table formatting for section 7 (#433), and ensuring the traceability between the V&V Plan and V&V Report aligned and matched up with one another (#434).

Source	**Issue #**	**Issue Description**	**Changes Made**	**Commit**
TA Feedback	#432	The document would benefit greatly from having additional comparisons between expected output and actual output to put the results in a more meaningful context	The actual result column has been updated to have a more meaningful comparison to the expected output result and provide more detailed context of the actual result	4353716

TA Feedback	#433	The tables in section 7 all need to be fixed; the indentation, justification and alignment when compared to the rest of the document makes it seem unprofessional and should be fixed	Fixed formatting of the tables	4353716
TA Feedback	#434	Traceability between the V&V Plan and V&V Report is inconsistent	Improved traceability between the two documents to ensure alignment and consistency across test cases	e5f3a37

Table 5: Feedback and changes for Verification and Validation Report

3 Extras

The extras set out for the system consist of both a Design Thinking Document and a Usability Report. The Design Thinking Document aims to explain the process behind, and work done towards, the final design of the system. This document explains the overall flow of processing through the system and the interfacing elements, external libraries utilized, and more. The document flow is iterative, as is the design process, and aims to communicate the entire design process of the system. The Design Thinking Document can be found at docs/Extras/DesignThinking/DesignThinkingWorksheet.pdf.

The Usability Report aims to focus on evaluating the system from a user perspective. This report documents usability testing conducted with participants to assess ease of navigation, clarity of instructions, and overall user experience. It includes observations, user feedback, and analysis of interaction patterns to identify the key areas to focus on and address. The findings are used to ensure the system meets the non-functional requirements and to guide improvements to the interface for both patients and physiotherapists. The Usability Report can be found at docs/Extras/UsabilityTesting/UsabilityReport.pdf.

4 Design Iteration (LO11 (PrototypeIterate))

4.1 Initial Design

Initially, the system was planned to be a patient-side only application that would support tracking and recording of patients exercises at home to ensure proper form for recovery. However, after several meetings with a physiotherapist advisor, it was determined that this initial design leaned into the grey-area of providing healthcare advice, something that the system nor its developers are entitled to do. It also seemed that the advisor would be interested in seeing the patients results from recordings.

4.2 Rev-1

The application was changed to support two-sided interactions from both patients and physiotherapists, however completely asynchronously. During this phase, the application was only meant to support patients recording their exercises which were prescribed by their physiotherapists. However, due to requirements to interact with the MediaPipe and OpenCV APIs, there had to be a data structure that identifies which landmarks on the patient's body are to be tracked. So, the system limited scope to focus on the knee and locked three landmarks for those measurements: the hip, knee, and ankle. With these three landmarks, the angle across the knee was attainable. There was a basic data structure created for knee-related exercises which stores the exercise instructions so they may be modified later, but this wasn't implemented until Rev1.

At this point the application is two-sided for both patients and physiotherapists, it supports recording and data gathering on knee-related exercises, but it didn't feel secure enough. To add a layer of validity for both physiotherapists and patients, physios are required to enter their physiotherapist identification number (something given to all physiotherapists once they pass an accredited physiotherapy program) and patients are required to enter an invite code (given to them by a physiotherapist with a valid account).

Combining the front-end in React to the backend written in Python required a roundabout approach to run .JSON responses across a Flask server. This way the back-end interfacing with the MediaPipe and OpenCV can pass the results of the video-feed analysis to the front-end which handles storing activity data in the database and displaying feedback to the patient on the front-end.

4.3 Rev0

After more meetings with the advisor and a meeting with the professor and TA, there was a desire for more communication between patients and physios. This brought about the creation of the feedback page which allows both patients and physios to leave comments on specific exercises if there is a need to address anything not captured by the application. There was also a basic profile page implemented that held contact information for both the patient and physiotherapist if another in-person session had to be booked or a more synchronous form of contact was required.

The progress page introduced a basic graphical overview of patient activities and repetitions done over the last week. These two fields were intended to support physio decision making when it come to exercise instruction modifications or planning another visit in-person. This page is intended to be a quick, 'at a glance' update of patient progress.

Otherwise, Rev0 time was spent implementing all of what was defined above to prepare for the Rev0 demo.

4.4 Rev1

Once there was a semblance of a finished product, it was easier to see what was lacking for the advisor and patients. The physio advisors requested more user-entered data rather than simply the computer vision gathered data, as well as specific data entry fields for sets and reps. Once exercise modification was implemented so physios could alter exercise instructions and add any level of detail, the two extra fields for sets and reps were added in the exercise modification modal to support expectation setting for both physios and patients during exercise recording and feedback reviewing. For the additional user-entered data, a survey was implemented that appears after each activity recording and prompts the user to rate each of the following on a scale from 0 to 10: pain or discomfort, level of effort, and satisfaction with the activity. Using these fields in combination with the gathered information from the system, the physio advisors felt more confident in their abilities to properly assess if exercise instructions or sets and/or reps required changing after an activity. These fields were also added to the progress page as graphical options so the physio can see trends in the user's entered data over the previous week.

After usability testing with several users and physios, there was a clear need for some usability tweaks and an explanation of several features. Patients were clearly not aware of all the options available to them on several screens and physios found the experience jarring or disconnected. The interface was modified to have more contrast between elements, and visual indicators such as borders and persistent scrollbars were added to clearly indicate the capabilities of all elements on the UI. There was also additional helper text added to each page to describe at a basic level the purpose of those screens. Helper text was added for certain cases, such as having no assigned exercises or no historical activity data on both the main dashboard and the feedback page.

Also, for the sake of user and physio convenience there was a disable option added to patient and physio accounts. Physios can disable linked patient accounts if they feel it is no longer required or if the patient has been inactive for some time. This can help the physio clean up their main dashboard and focus on users that are actively using the system for their recovery. For patients, if they feel the system isn't supporting them as well as they had hoped, or they feel they have made adequate recovery, they can disable their own account to stop receiving feedback and recording requests from their physio. Several other UI modifications, such as logout capabilities on the profile page as well as main page and linked physio contact information display were implemented to increase quality of life for both patients and physios.

To prove the scalability of the design, exercise enabling and disabling was also added to the physio side such that individual patients could have different assigned exercises with personalized instructions, sets, and reps. With simply a checkbox and a 'modify instructions' button placed on each exercise, the physio could completely personalize a patient's capabilities within the system. This is the final update that was shown during Rev1 final demonstration and at the Engineering Expo on April 7th, 2026.

5 Design Decisions (LO12)

5.1 Limitations

After the Rev0 demonstration, the system was locked to using one camera for pose detection rather than 2 from different angles. This limitation was placed on the system for both ease of implementation as well as user convenience. Under an assumption, every user would have a camera accessible to them on their smart device, however the need for an additional camera which would have to link to the device and be separately identified by the system would cause many changes in the backend implementation and place more responsibility and requirements on the user. Additionally, during the PoC demo, it was determined with the professor that one camera would be enough for pose detection and form correction. This made the implementation for pose detection and form correction much simpler for the team, a far more manageable workload for such a project.

Once the back-end and front-end were linked with a Flask server to pass responses back and forth, the performance of the pose detection suffered drastically. Ultimately unaware of how to improve the performance and whether it was an implementation issue, a design issue, or perhaps an issue of stack selection, the system moved forward and assumed a slower rate of movement for patient recording which increased the accuracy of the pose detection even with such high latency from the Flask response passing. There was a thought that changing the front-end to a python supportive language could reduce latency, however it was discovered so late into the project and with no guarantee that it would solve latency and a definite guarantee of needing to re-implement the entire front-end of the system, this limitation of pose detection latency was accepted as an overall system performance limitation. If there was more time devoted to this project, it would be a good idea to rebuild it from the ground up using a python integrated front-end language to attempt to limit pose detection latency.

5.2 Assumptions

It was assumed that patients would only have injuries and limitations related to the knee area, and that users would perform exercises which utilized knee flexion and extension to improve range of motion and strength. This was assumed for the simplicity of implementation for recording and pose detection, and to allow movement along a single plane which would be observable by a single camera. The simplicity of implementation reduced the requirements for the exercise data structure, since the MediaPipe and OpenCV APIs were told to only focus on the hip, knee, and ankle joint locations there was no need to store the landmarks to be tracked in the exercise. This alteration to the data structure would have required a rewriting of all 5 video handling modules and a more intricate data passing of the exercise data structure between these modules to ensure all relevant information was tracked. The single plane movement allowed an ease of use for patients recording their exercises, since all they must do is bend and/or extend their knee along the plane perpendicular to the camera.

User honesty was assumed for patient entered data, that is pain, effort, and satisfaction with the activities. These fields cannot be determined using computer vision as they are all internal and every user handles them differently. Even when interacting with physiotherapists in their office, these data points are still relying on the user's honesty. Since these fields and their data would be useless to the physiotherapist if they weren't answered honestly, and therefore irrelevant and unhelpful in the patient's recovery, the system assumes honesty from the patient to assist in their recovery.

A stable internet connection was assumed from the beginning of implementation to simplify data handling. With a stable internet connection, data can always flow between the database and the user's device, this created no need for offline backups or delayed uploads to the database. For example, if a user finished an exercise but couldn't connect to the internet that data should be stored and uploaded when they have internet access again. However, this created an issue as the team was unsure how to implement this and it wasn't related to the overall idea for the application. This assumption simplified the implementation and narrowed scope to a more patient and physio focused system rather than a data management system that could handle all edge cases.

5.3 Constraints

Financial constraints placed on the project kept the system limited to using free APIs and cloud database services for implementation. Usage of MediaPipe and OpenCV's APIs is publicly available and was enough to support the creation of the system up to the final Rev1 iteration. Google's Firebase was used as a cloud database system, specifically the Firestore database was used to store user, exercise, and activity data. The limited size of the free Firebase meant we excluded video storage from our solution. Initially, there was intention to draw live feedback from the video-feed and further data collection from the recording after the user had submitted their video, however the size limitation of the cloud database meant the system couldn't store all that video footage. Instead, both data and feedback are compiled during the live video-feed, and the data is simply saved to the database after recording is complete.

There was a constraint discovered early for healthcare advice. As the system developed by software engineers is not capable of offering healthcare advice or altering patient exercise programs, this drastically altered the iterations after the proof-of-concept demo. There were initial plans to gamify exercises, allowing users to unlock different exercises by doing enough of their first assigned exercises, or perhaps to switch exercises with an earned in-application currency to increase user interaction and consistency in using the system. However, this had to be abandoned and ultimately drove the application to become the two-sided system embodied in the final iteration. This way, physios can provide healthcare advice through the system and there is no responsibility of instruction or set and rep management done by the application. This constraint heavily impacted the direction that this project went it, limiting the system's ability to autonomously make exercise decisions or alterations, however the final two-sided resulting application feels more alive with interactions between physios and patients.

6 Economic Considerations (LO23)

Yes, there is a clear market for our product. Existing physiotherapy companion apps such as Physiapp and Physiomobile offer progress tracking (checklist-based) and gamification (streaks), but they lack real-time exercise form feedback. Our app fills this gap by using OpenCV and MediaPipe pose landmarks to analyze patient movement during home exercises, giving immediate corrective feedback. Additionally, we provide value back to the physiotherapist by automatically sharing tracked metrics (e.g., reps, form quality) together with subjective patient data (pain levels, perceived exertion). This addresses the gap for patients and physios between appointments.

Marketing Approach

Marketing would target physiotherapy clinics and independent practitioners. Strategies include:

- Direct outreach to private clinics via email and informational webinars.
- Free trial periods for clinics to demonstrate improved patient adherence and outcomes.
- Testimonials and case studies from early-adopter clinics.

Cost to Produce Application

We estimate the cost to develop our application to a production-ready version (iOS/Android app, backend dashboard for physios, pose-analysis pipeline) at:

- Development (4 months, 2-3 engineers): $60,000 - $80,000.
- Exercise pose analysis expertise (consultation with physiotherapists to validate exercise models and feedback logic): $10,000 - $15,000.
- Cloud infrastructure (AWS/GCP for pose inference): $5,000 initial setup + $500/month.
- Legal/regulatory (privacy, medical device classification, liability for recording and providing feedback): $15,000 - $20,000
- Marketing launch: $5,000.

Total initial cost: approximately $95,000 - $125,000.

Pricing Model

We would not charge individual patients. Instead, we adopt a subscription model for physiotherapy clinics, tiered by the number of active patients per month. A "small private clinic" typically has 2-3 physiotherapists, each seeing 25-50 patients per week. Therefore, reasonable patient tiers are:

- **Tier 1 (up to 50 active patients / month):** $50 / month
- **Tier 2 (up to 150 active patients / month):** $100 / month
- **Tier 3 (unlimited patients):** $200 / month

Breakeven Analysis

Assume an average subscription of $100 / month per clinic (weighted toward Tier 2). Monthly recurring costs (cloud, support) are roughly $500 for every 100 clinics. Development cost is a one-time $110,000.

Monthly profit per clinic after variable costs: $100 - $5 = $95. Number of clinic-months needed to recover development cost:

$$\frac{\$110,000}{\$95 \text{ per clinic-month}} \approx 1,158 \text{ clinic-months.}$$

If we acquire 100 clinics in the first year, breakeven occurs after about 11.6 months. In terms of total clinics, we would need roughly 97 clinics subscribing for one full year to break even.

Potential Users (TAM / SOM)

- **Total Addressable Market (TAM):** All physiotherapists in Canada - approximately 22,000 registered physiotherapists (Canadian Institute for Health Information). Assuming an average of 5 physiotherapists per private clinic (typical for a medium-sized clinic), that gives roughly 4,400 clinics.
- **Serviceable Available Market (SAM):** Private clinics in Ontario, where our initial launch would focus. Ontario has roughly 6,000 physiotherapists, translating to about 1,200 private clinics.
- **Serviceable Obtainable Market (SOM):** In the first two years, with targeted marketing, we could realistically capture 1% of the Ontario market - about 12 clinics. That would generate monthly revenue of $1,200 (at $100 average). With successful promotion and marketing of the application, the application could grow to 5% (60 clinics) after three years, achieving monthly revenue of $6,000 and approaching a breakeven point.

7 Reflection on Project Management (LO24)

7.1 How Does Your Project Management Compare to Your Development Plan

The team had largely followed our initial development plan. Although the first version did not include clear details in the early stages, when we revisited the document later, we were able to retain most of its structure. We stuck to our team meeting plan and team communication plan by maintaining open lines of communication among team members and reflecting after each deliverable on what went well and what could be improved.

The workflow plan was helpful for us and had defining team roles assisted us in the division of responsibilities effectively. However, we recognized that the roles would have been even more useful if we had reflected on them after each deliverable, rather than only at the beginning.

One area that we would like to improve on was our use of technology. While we had placed importance on implementing a CI/CD pipeline, due to the way we divided work, it became difficult to prioritize tools such as GitHub Actions or the GitHub Projects tab later in development.

7.2 What Went Well?

In terms of what went well throughout our project, the team's overall organization and collaboration is one of the first few things that come to mind. We utilized GitHub Issues and version control to stay on track of all our deliverables, tasks, and progress. This ultimately allowed us to stay aligned and ensure everyone was held accountable for their assigned tasks. We also clearly divided up the respective responsibilities between the frontend and backend components to ensure everything remained as efficient as possible. Moreover, we successfully implemented real-time feedback, one of the core features of the app. We were able to successfully process frames continuously and return the relevant metrics (e.g., knee angle, rep count, ROM). This took numerous iterations to refine but we were so happy to see everything starting to finally come together.

7.3 What Went Wrong?

One of the main issues we faced was handling the accuracy of the pose detection and real-time motion analysis. Accurately detecting knee movement within a live environment was more difficult than expected, especially when dealing with inconsistent lighting, camera positioning, and partial visibility of the user. In addition to that, another issue we kept coming across was the accuracy of repetition counting. Any small variations in movement or added noise during the pose detection sometimes caused missed, delayed or incorrect rep counts. This required multiple iterations of threshold tuning and logic refinement, which took more time than initially planned. We also encountered numerous issues during the frame processing integration. Since the system relied on the backend for continuous frame processing, this sometimes led to performance issues such as latency or delayed feedback, which posed as a key limitation for us. Looking back, some of these technical challenges caused extreme delays and even though we continued to communicate as a team through these hard times, this did affect our pre-planned development schedule. However, regardless of these challenges, this whole experience including both the successes and challenges was all around a valuable learning experience for all of us, one that we will continue to hold very close to our hearts and are grateful for. Moreover, it genuinely helped us better understand the challenges of building real-time, user-based applications, which ultimately has better prepared us for our futures in our post-undergraduate careers.

7.4 What Would you Do Differently Next Time?

Moving forward, we would like to first place a stronger emphasis on early prototyping and the testing of our features. There was a challenge with the pose detection which was more apparent towards the end of development. Initially we had done our testing on python only but when trying to integrate it into the backend, we had to make a lot of adjustments and changes which caused a lot of delays. If we had done more testing earlier on in development, we would have been able to identify these issues sooner and make the necessary adjustments much earlier on.

We also would like to place a stronger emphasis on the use of tools such as GitHub Projects and GitHub Actions to help with our project management and workflow. We had initially planned to use these tools but due to the way we divided work, it became difficult to prioritize them later on in development. If we had placed more importance on these tools from the beginning, we would have been able to better track our progress and automate some of our processes, which would have ultimately made our development process more efficient.

Finally we would build in time for each team member's schedules and abilities. We had a lot of technical challenges that caused delays and we had to adjust our schedule accordingly. If we had built in more time for these types of challenges, we would have been able to better manage our schedule and avoid some of the stress that came with trying to meet deadlines while also dealing with unexpected issues.

8 Ethics and Equity (LO18)

When the team initially formed, the project requested was simply “something to help those in need”. After discussing and finding common ground the team decided to create this physiotherapy companion application to help recovery and consistent healing of physiotherapy patients. Advisors and usability participants were selected and thoroughly questioned to ensure usage of the system was easy and helpful to average patients. All communications systems were tested thoroughly to ensure patients could be in contact with their physios through the system, and external contact methods were stored and presented within the profile page to ensure additional meetings and synchronous contact could occur if required.

Additionally, data collection from users of the system and usability participants was kept as limited as possible to ensure privacy. Names and emails were omitted where possible to ensure privacy within our documents for usability participants. For system users, video feeds were not stored for both data limitations with the cloud database and also personal data privacy. This way there is no video or photography of users to be leaked if a data breach should occur.

Finally, we thanked our physio advisors and usability test participants extensively for their help in the creation and refinement of our labour of love.

9 Reflection on Capstone

9.1 Which Courses Were Relevant

Maham
SFWRENG 2AA4 - Software Design 1 introduced me to viewing software development through a more holistic lens. It raised my awareness of and emphasized many important areas such as what an MIS is, what software design principles are, and how to define software components mathematically.
SFWRENG 2DM3 and 2FA3 - Discrete Math 1 and 2 equipped me with the ability to translate natural language to mathematics. Given the MIS's extensive use of discrete math, proficiency in this area was imperative.
SFWRENG 3A04 - Large System Design introduced software architectural approaches to development. As part of a team, I had to develop a software and it was very similar to what was done in capstone, but at a significantly faster pace. This gave me valuable insight into what capstone entails.
SFWRENG 3RA3 - Requirements was essential to learning how to construct requirements for software that fulfills software quality benchmarks while also being tailored for the correct audience.
SFWRENG 3S03 - Software testing provided a concise introduction to software testing which was fundamental to deliver a functional product by the end of development.
SFWRENG 4HC3 - Human computer interfaces was a very important course, since it emphasized designing software with a human-centered approach. It taught us develpment priorities and the importance of always keeping the audience in mind.

Vaisnavi
Requirements Engineering (3RA3)

- Taught us how to engineer and refine system requirements.

- Helped significantly in defining the functional and non-functional requirements

Software Architecture (3A04)

- Taught us how to design larger systems as independent, modular components, which allowed for separation of concerns within our various modules

Software Testing (3S03)

- Taught us how to conduct both unit and integration testing.
- Helped us verify the reliability of key features like pose detection and repetition counting.

Software Development (2AA4 & 2C03)

- Taught us the fundamentals of software development and best practices for building scalable systems.

Human Computer Interfaces (4HC3)

- Taught and provided us with the correct tools to design our user interface, ensuring it was user-centered and intuitive.
- This course also provided us with the knowledge and tools to allow us to successfully conduct usability testing and refine our mobile application based on the user feedback.

Matthew
Throughout this project, many courses were relevant to our capstone project. More specifically for the documentation, several of my communication and documentation related courses touched on how to effectively communicate the software specifications. For example, **SFWRENG 3A04 - Large System Design**, where I had to follow and write software specification. Furthermore, with the coding aspect, I applied learning from **SFWRENG 3S03 - Software Testing** to help write the unit tests. **SFWRENG 4HC3 - Human Computer Interfaces**, helped with how to design with human compatibility as well as how to conduct usability tests for our application effectively. Finally, the design of databases uses practices from **SFWRENG 3DB3 - Databases** and **SFWRENG 2OP3 - Object Oriented Programming** was used to help for the design of the backend Python code. Overall, the application of the courses helped us throughout the capstone project.

9.2 Cieran

SFWRENG 2AA4, 3BB4, 3A04 – Software Development

- Extensive knowledge and practice with overall system and module development.
- Breakdown of overall systems into modules with responsibilities and avoidance of circular dependencies.
- \- General logical thinking for system design with an emphasis on scalability, extendibility, and modularity.

SFWRENG 3DB3 – Databases

- Proper usage of databases, queries, joins, and data types.
- Proper fields included in database tables, necessary key fields and foreign keys to link tables to each other.

- Aim to reduce redundancy of data, avoid storing and needing to update data in multiple places.

SFWRENG 3SO3 – Software Testing

- Unit, usability, functional requirement, and non-functional requirement testing suite development.
- Overall focus on code coverage, branch navigation and coverage, and edge case assurances.
- Helped with learning new test suite development and working to failsafe developed functions, classes, interfaces, and modules.

SFWRENG 4HC3 – Human Computer Interfaces

- Development of front-end interfaces for both patient and physio sides of the application.
- Emphasis on understandability and learning of system functionality using visual indicators and assistance text or other materials where required.

9.3 Knowledge/Skills Outside of Courses

Vaisnavi
One of the most significant areas where we had to do the most learning was for our recording feature of the app, specifically the pose detection and computer vision logic. We used OpenCV and MediaPipe to process the images in our backend, allowing us to display and extract key landmarks and compute the relevant knee angles. However, learning and working with these technologies at the same time was very new and extremely challenging. There were so many different aspects to it and we struggled a lot as mentioned previously with latency. We realized potential better methods to reduce the latency and increase the accuracy a bit too late into the project where it was too late to change everything, but it was a valuable learning experience overall. We learned a lot in terms of adapting to a new skill in a specified timeline and we are super grateful for this opportunity to make the mistakes that we did, but also learn from those mistakes at the same time.

Matthew
Throughout this project, I needed to learn skills outside of my course work. Some of the skills that I needed to learn was working with computer vision and how the pose detection worked. In order to learn these skills, I had to look through the documentation as well as work through tutorials to then try and apply it to the problem of our project. Furthermore, when working on some of the documents for our project, I had to refer to previous course work but also how it is written.

Maham
Several important lessons emerged throughout this project. From a technical perspective, I learned to use React after determining that only the analysis logic should reside in the backend. Additionally, I experimented with database logic and sought to seamlessly integrate both components into a cohesive system. I also recognized the importance of maintaining clear, thorough documentation at each stage of development. This process allowed me to refine my ability to communicate concisely while still providing sufficient detail. This skill proved to be closely tied to my time management abilities. Time management is a lifelong pursuit and a skill that is developed

through practice and experience. As a communicator, I acknowledge that I tend to be somewhat overcommunicative. I had to temper my approach to synchronize with the styles of my teammates and ensure that I was not overwhelming them. It was a balancing act between providing too little and too much, but in the end, I managed to successfully balance our styles. Similarly, I had to learn to navigate different work dynamics. I realize that I may have initially set out to be inflexible because I did not want anyone's work to be contingent on mine, but I ultimately delivered on time (mostly). Even when I did not, my teammates were gracious enough to offer assistance and additional time when I needed it.

Cieran

During this project I learned how to apply several API interfaces from MediaPipe, OpenCV, and Firebase. Working with these really made my systems knowledge feel far more extensive than it had in the past, and far more confident about the range of capabilities my systems could accomplish. I had never used React for any extensive project before, so learning all the capabilities and implementing a full front-end of a system was a very interesting educational experience. Working with an actual database, in the form of Firebase, was a very interesting opportunity to use my database knowledge from 3DB3, however it also highlighted how logical the process is working with a database and how systematic thinking has shaped the way I perceive and imagine making and manipulating database tables. Learning Jest, a type script testing suite, was intuitive but also almost a fun exercise in code coverage and branch management within functions and classes. Overall I feel the knowledge I gained outside of courses to advance this project have increased my skill and confidence as a software engineer.